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Saveetha Institute of Medical and Technical Sciences

Chennai–602105

A CAPSTONE PROJECT REPORT

EARLY FOREST FIRE PREDICTION AND ALERT SYSTEM WITH HEAT MAP-BASED WILDLIFE AND VEGETATION PROTECTION

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

CSA1109 – Object Oriented Analysis and Design

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN COMPUTER SCIENCE AND ENGINEERING

Submitted by

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Under the Supervision of

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SEPTEMBER 2026

Saveetha Institute of Medical and Technical Sciences
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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**Early Forest Fire Prediction and Alert System with Heat Map-Based Wildlife and Vegetation Protection**" has been carried out by **Chethrika Sri (Reg. No. 192511112)** under the supervision of **Dr. Mary Valentina** and is submitted in partial fulfilment of the requirements for the current semester at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Chethrika Sri (192511112) , hereby declare that the Capstone Project Work entitled “Early Forest Fire Prediction and Alert System with Heat Map-Based Wildlife and Vegetation Protection” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that applicable ethical, academic, institutional, and professional requirements have been followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 08/09/26

Signature of the Students with Names

Chethrika Sri (192511112)

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Chethrika Sri / 192511112	Module 1 – User Authentication & Forest Monitoring; Module 3 – Priority-Based Heat Map Generation; system integration support.	Design of secure access/monitoring workflow and heat-map priority logic.	Review of environmental-data flow, risk classification and priority assignment requirements.	Chapters 1, 3, 5 and assigned module sections.	60%
Kamali R / 192521217	Module 2 – Forest Fire Prediction; prediction workflow and related documentation.	Design of threshold/historical-pattern prediction workflow.	Review of risk classification and prediction validation requirements.	Chapter 2 and assigned prediction sections.	40%
Total					100%

ABSTRACT

Forest fires are a major environmental challenge because they can spread rapidly and cause irreversible damage to vegetation, wildlife, biodiversity, ecosystems, nearby communities, and property. The project addresses the limitations of conventional fire detection approaches that depend heavily on manual surveillance and may detect fires only after ignition, leading to delayed response. The proposed Early Forest Fire Prediction and Alert System with Heat Map-Based Wildlife and Vegetation Protection uses real-time environmental information including temperature, smoke level, humidity, weather conditions, and GPS location to identify abnormal conditions and predict potential fire risk at an early stage. The system is organized into four modules: User Authentication & Forest Monitoring, Forest Fire Prediction, Priority-Based Heat Map Generation, and Alert & Emergency Notification. The prediction module compares sensor values with predefined thresholds and considers historical forest-fire patterns and current environmental conditions to classify risk as Low, Medium, High, or Critical. The heat-map module combines fire-risk information with wildlife density and vegetation coverage to prioritize ecological zones for response. High and Critical risk conditions trigger notifications to authorized forest officials containing location and risk information. The design emphasizes secure access, real-time monitoring, risk prioritization, traceability, and rapid emergency communication. The project supports Sustainable Development Goal 15 – Life on Land by focusing on protection of forests, wildlife, and vegetation. The supplied presentation does not provide quantitative deployment or test measurements, so this report distinguishes planned verification from measured results and does not invent numerical performance values. Future development proposed by the project includes satellite and drone integration, advanced AI models for fire-spread prediction, a mobile application, automatic emergency response and resource allocation, and expansion to larger forest regions.

Keywords: Forest Fire Prediction, Heat Map, Wildlife Protection, Real-Time Monitoring, GPS Alerts, SDG 15

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
GPS	Global Positioning System	—
SMS	Short Message Service	—
AI	Artificial Intelligence	—
UML	Unified Modeling Language	—
SDG	Sustainable Development Goal	—
OTP	One-Time Password	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Forests play a vital role in maintaining ecological balance by supporting biodiversity, regulating climate, and providing essential natural resources. The submitted project identifies forest fires as an increasing environmental challenge that can cause extensive damage to vegetation, wildlife, and ecosystems while threatening human lives and property. Conventional detection methods based on manual surveillance can delay detection and emergency response. The proposed system addresses this challenge by monitoring temperature, smoke, humidity, weather conditions, and GPS location and using these inputs to identify potential fire hazards at an early stage.

1.2 Need for the Project

- Forest fires spread rapidly and can cause irreversible environmental and ecological damage.
- Existing approaches may detect fires only after ignition, reducing the time available for intervention.
- Forest officials need timely information about the location and severity of potential fires.
- Ecologically sensitive areas with dense wildlife and vegetation require priority-based response.
- Real-time alerts with GPS information can support faster and more informed emergency action.

1.3 Problem Statement

Forest fires spread rapidly, causing irreversible damage to forests, wildlife, vegetation, biodiversity, and nearby communities. Existing systems primarily detect fires after ignition, resulting in delayed emergency response and increased environmental loss. There is a lack of an integrated system that can predict forest fires early, assess their severity, prioritize high-risk ecological zones, and provide real-time alerts with precise GPS locations. Therefore, an intelligent early prediction and alert system is required to enable timely intervention, minimize damage, and protect forests, wildlife, and vegetation effectively.

1.4 Project Aim

To design an integrated system that predicts forest-fire risk at an early stage, prioritizes ecologically sensitive zones using a heat map, and delivers immediate GPS-based alerts to authorized forest officials.

1.5 Project Objectives

1. To predict forest fires at an early stage using real-time environmental data.

2. To generate a priority-based heat map for effective protection of wildlife and vegetation.

3. To provide instant alerts with GPS location to authorized forest officials for rapid response.

1.6 Scope

Included: secure user authentication; real-time forest monitoring; collection of temperature, smoke, humidity, weather and GPS data; threshold and historical-pattern analysis;

Low/Medium/High/Critical risk classification; priority-based heat-map generation; SMS/application notifications to authorized officials; alert history.

Excluded from the current scope: satellite and drone data integration, advanced AI fire-spread prediction, mobile application development, automatic emergency response/resource allocation, and large-scale regional expansion. These are explicitly identified as future enhancements.

Assumptions: environmental sensors provide usable readings; GPS information is available; authorized officials have access to the alert channel; and historical fire-pattern information is available for analysis.

Boundary conditions: the current design focuses on early prediction and alerting and does not claim autonomous firefighting or complete fire-spread simulation.

1.7 Expected Engineering Outcome

The intended outcome is an integrated monitoring, prediction, prioritization and alerting system. It should provide secure access, reliable environmental data, early fire-risk classification, an ecological-priority heat map, and rapid notifications containing location and fire-risk information.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The submitted project presentation provides references covering UML-based analysis and design, software engineering, Java documentation, MySQL documentation, and Sustainable Development Goal 15. These references were used as the source basis for the design-oriented discussion in this report. The PPT does not provide a detailed systematic literature-search protocol or a comparative survey of commercial fire-detection products.

2.2 Review of Existing Work

Source	Relevant Work / Topic	Relevance to Project
Booch, Rumbaugh & Jacobson (2005)	The Unified Modeling Language User Guide	Supports UML-based analysis and system modelling.
Pressman & Maxim (2019)	Software Engineering: A Practitioner's Approach	Provides software engineering and development principles relevant to system design.
Oracle Corporation (2025)	Java Platform, Standard Edition Documentation	Referenced implementation technology documentation; exact implementation details are not shown in the PPT.
Oracle Corporation (2025)	MySQL 8.0 Reference Manual	Referenced database documentation; exact database implementation details are not shown in the PPT.
United Nations (2025)	Sustainable Development Goal 15: Life on Land	Provides the sustainability context explicitly linked to the project.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual surveillance	Direct human observation; familiar workflow	Delayed detection; slower emergency response	Motivates continuous automated monitoring.
Detection after ignition	Can identify confirmed fire conditions	May miss the earliest warning window	Proposed system focuses on prediction before/around early ignition conditions.
Integrated early prediction + priority heat map + alerts	Combines risk prediction, ecological prioritization and rapid communication	Requires reliable sensors, data and connectivity	Directly matches the proposed project architecture.

2.4 Research / Engineering Gap

The problem statement in the PPT identifies a gap in systems that integrate early prediction, severity assessment, ecological-zone prioritization and real-time GPS alerts. The proposed design addresses this gap by linking environmental monitoring to risk classification, then combining risk with wildlife density and vegetation coverage to assign response priority.

2.5 Proposed Contribution

The proposed contribution is a four-module integrated system in which secure forest monitoring supplies environmental data, prediction classifies fire risk, the heat-map module prioritizes sensitive ecological areas, and the alert module communicates actionable information to authorized officials. The approach is intended to reduce response delay and improve allocation of emergency attention.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Forest Officials	Receive High/Critical risk alerts with GPS location, temperature, smoke level, risk and priority information.
Authorized System Users	Securely log in and access monitoring information.
Forest Monitoring Team	Obtain continuous environmental readings including temperature, smoke, humidity, weather and GPS.
Emergency Response Personnel	Identify priority ecological zones using the heat map and allocate response attention accordingly.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Secure user authentication	Functional	Only authorized users can access the system.
Real-time environmental monitoring	Functional	Temperature, smoke, humidity, weather and GPS data are collected and passed to prediction.
Early fire-risk prediction	Functional	Classify conditions as Low, Medium, High or Critical.
Priority heat-map generation	Functional	Assign Priority 1–4 using risk, wildlife density and vegetation coverage.
Emergency notification	Functional	Generate notifications for High/Critical risk with location and risk details.
Alert history	Functional	Maintain records of generated alerts.
Reliability	Non-Functional	Useful monitoring subject to sensor/network availability.
Security	Non-Functional	Authentication and authorized-user restrictions are applied.

3.2 Constraints & Applicable Standards

The PPT does not explicitly claim compliance with a particular regulatory standard. The following are therefore treated as technical references rather than unsupported compliance claims.

Standard / Code / Technical Reference	Requirement	How Applied in This Project
UML (version not specified in PPT)	Model system structure and interactions	UML diagram is included in the project presentation.

Java Platform, Standard Edition Documentation (Oracle, 2025)	Reference for Java-based software development	Listed in project references; exact usage is not shown.
MySQL 8.0 Reference Manual (Oracle, 2025)	Reference for relational data management	Listed in project references; exact schema is not shown.
UN SDG 15 – Life on Land	Protect terrestrial ecosystems and biodiversity	Project explicitly links its environmental purpose to SDG 15.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Environmental inputs	Temperature, smoke, humidity, weather and GPS	Sensor/location data	High
Risk classification	Low / Medium / High / Critical	Category	High
Heat-map priority	Priority 1–4	Category	High
GPS location	Precise location included in alert	Coordinates	High
Alert trigger	High or Critical risk	Condition	High
Notification recipients	Authorized forest officials only	Users	High
Alert history	Maintain historical alert records	Records	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual surveillance. This relies on human observation and reporting, which is simple but may delay detection and response.

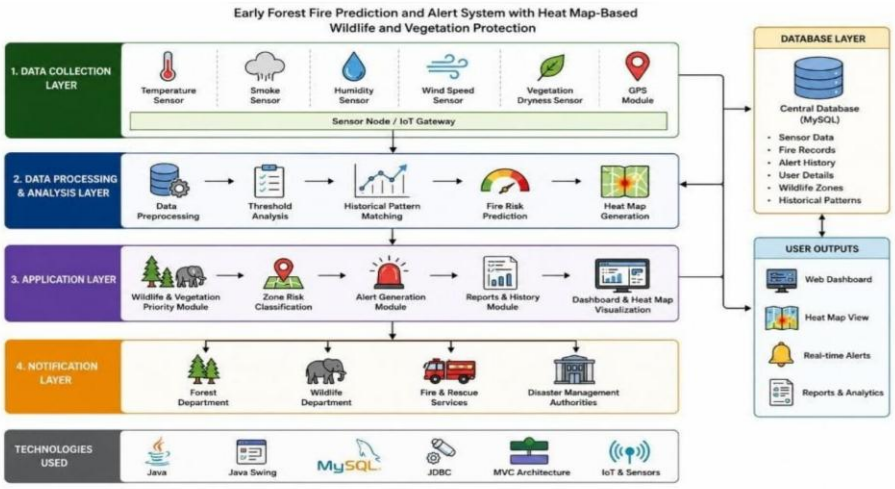
Alternative 2: Sensor-based early prediction with priority heat map and alerts. This integrates the four proposed modules and directly addresses early detection, ecological prioritization and communication.

Alternative 3: Future multi-source system using satellite/drone data and advanced AI. The PPT identifies these as future enhancements rather than current scope.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	Low	High	High
Cost	20%	Low	Medium	High
Safety / response	20%	Low	High	High
Sustainability	15%	Medium	High	High
Reliability / scalability	15%	Low–Medium	Medium–High	High

3.5 Selected Design & Detailed Design

Alternative 2 is selected because it directly implements the objectives stated in the PPT while remaining within the current project scope. It combines real-time environmental monitoring, early risk classification, ecological prioritization and immediate alerts without depending on future satellite, drone, mobile or advanced-AI capabilities.



System-level interaction: Module 1 authenticates users and supplies environmental observations. Module 2 analyses current sensor conditions and historical patterns to classify fire risk. Module 3 combines predicted risk with wildlife density and vegetation coverage to generate response priorities. Module 4 communicates High/Critical conditions to authorized officials with GPS and environmental details and stores alert history.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Threshold + historical-pattern prediction	Threshold-only detection	Reduces dependence on a single condition and can reduce false alarms	Requires historical data	Selected because the PPT explicitly combines thresholds and historical patterns.
Priority-based heat map	Risk map without ecological weighting	Protects high-risk, ecologically sensitive zones first	Requires wildlife/vegetation information	Selected because wildlife density and vegetation coverage are core to the objective.
Authorized alerts	Alerts to all users	Improves security and relevance	Requires user authorization	Selected because the PPT specifies authorized forest officials.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Protection of forests, wildlife, vegetation and biodiversity	Project purpose and explicit SDG 15 linkage	Environmental protection is a primary design objective.
Ethics	Limit alerts/access to authorized personnel	Authorized-user requirement in Modules 1 and 4	Access control is included in the architecture.
Health & Safety	Rapid communication of High/Critical risk	Alert module and emergency-response purpose	GPS-based alerts support rapid action.
Security	Authentication and restricted notification recipients	Secure login and authorized-personnel notifications	Security is built into monitoring and alerting.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Sensor failure / poor sensor quality	Medium	High	High	Sensor validation, maintenance and cross-checking where available	Medium
Harsh weather affects sensing	Medium	Medium	Medium	Monitor sensor health and interpret readings with environmental context	Low–Medium
Network connectivity failure	Medium	High	High	Store local records where possible and retry/restore communication	Medium
False positive prediction	Medium	Medium	Medium	Use thresholds plus historical-pattern analysis	Low–Medium
Unauthorized access	Low–Medium	High	High	Secure login and authorized-user controls	Low–Medium
Incorrect GPS/location data	Low–Medium	High	High	Validate location data before alert generation	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

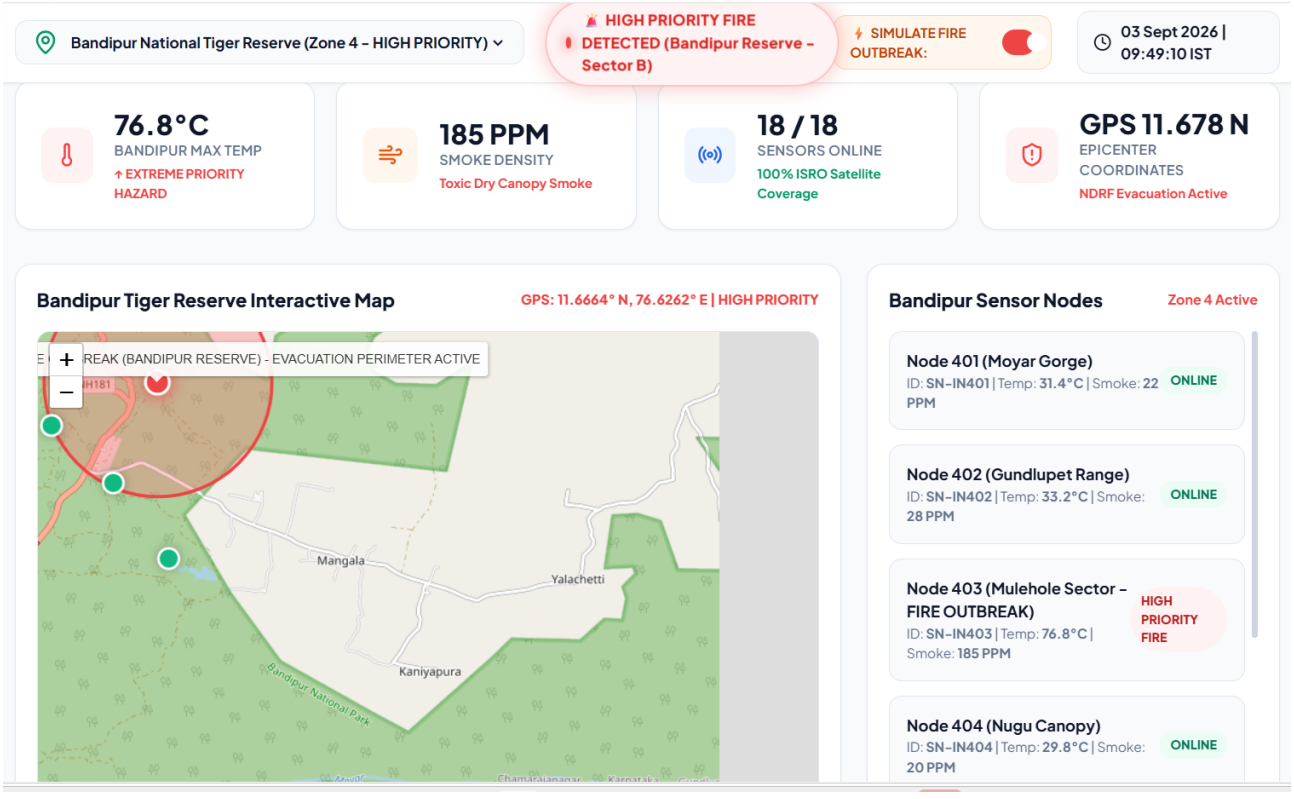
4.1 Development Approach & Resources

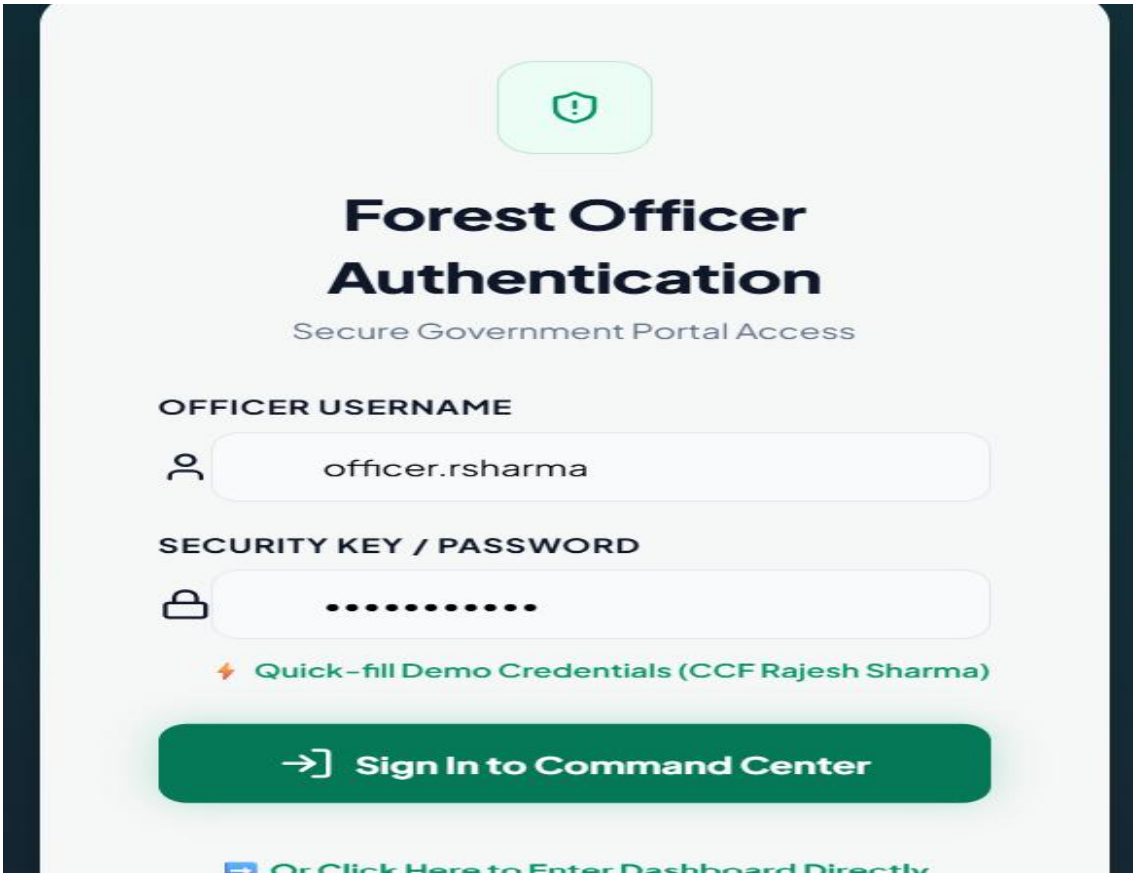
The system is structured using object-oriented analysis and design concepts and is divided into four functional modules. The design begins with secure forest monitoring, passes collected data to prediction, converts risk into ecological priority, and then triggers authorized emergency notifications. The project references Java and MySQL documentation, but the submitted PPT does not show implementation code, database schema, hardware specifications, or measured deployment results; these details are therefore not claimed beyond the evidence available.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Environmental sensors	Capture temperature, smoke, humidity and environmental conditions	Monitoring input; specific models not specified
GPS	Provide location for monitoring and alerts	Monitoring / alerting
Java Platform SE	Referenced software-development platform	Implementation reference; exact usage not shown
MySQL 8.0	Referenced relational database platform	Data-management reference; exact schema not shown
UML	System modelling	Analysis and design
SMS / application notifications	Communicate High/Critical alerts	Emergency notification design

IMPLEMENTATION





4.3 Test Plan & Verification

The PPT does not provide measured test results. The following is therefore a verification plan derived directly from the stated functions and objectives.

Requirement	Test Method	Acceptance Criterion
User authentication	Attempt authorized and unauthorized login	Only authorized users obtain access.
Environmental monitoring	Feed normal and abnormal environmental readings	Readings are collected and transferred to prediction.
Risk classification	Test readings below/near/above thresholds with historical context	System produces Low/Medium/High/Critical classification.
Heat-map priority	Provide combinations of risk, wildlife density and vegetation coverage	Priority 1–4 is assigned according to defined rules.
High/Critical alert	Trigger High and Critical risk cases	Alert is sent only to authorized officials and includes required details.
Alert history	Generate multiple alerts and inspect records	Alert history contains the generated events.

Verification status summary

Specification	Required Value	Achieved Value	Status
Four modules defined	4	4 modules described in PPT	Pass
Risk levels	Low/Medium/High/Critical	Explicitly defined in PPT	Pass
Heat-map priorities	Priority 1–4	Explicitly defined in PPT	Pass

GPS alert contents	Location + environmental/risk details	Explicitly specified in PPT	Pass
Quantitative performance	Measured results	Not reported in PPT	Not assessed
Field/deployment validation	Evidence required	Not reported in PPT	Not assessed

4.4 Results

The project presentation reports qualitative outcomes rather than numerical measurements. It states that the proposed system predicts forest fires at an early stage, provides real-time monitoring, generates a priority-based heat map, and sends instant GPS-based alerts to authorized officials. The conclusion states that these functions are intended to reduce forest-fire damage, improve emergency response, and protect wildlife and vegetation more effectively. No quantitative accuracy, response-time, false-alarm, sensor-performance, or field-deployment results are provided in the PPT.

4.5 Data Analysis & Engineering Judgment

Since no experimental dataset or measured performance results are supplied, quantitative statistical analysis cannot be supported from the source material. Engineering judgment is instead based on the stated design logic: combining threshold values with historical patterns is intended to reduce false alarms; ecological weighting is used to prioritize areas where wildlife and vegetation are dense; and GPS-based notifications provide actionable location information. The major dependencies are sensor quality, data availability, harsh weather conditions and network connectivity, which are explicitly listed as limitations in the PPT.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	**Evidence**	**Achievement (Fully / Partially / Not Achieved)**
Early prediction using real-time environmental data	Monitoring collects temperature, smoke, humidity, weather and GPS; prediction classifies risk.	Partially evidenced — no quantitative validation reported.
Priority-based heat map	Module 3 assigns Priority 1–4 using risk, wildlife density and vegetation coverage.	Partially evidenced — no field validation reported.
Instant GPS alerts to officials	Module 4 specifies alerts for High/Critical risk with GPS and risk details.	Partially evidenced — no delivery-time measurements reported.

4.7 Strengths & Limitations

Strengths:

- Early-risk prediction rather than relying only on post-ignition detection.

- Real-time environmental monitoring.
- Priority-based heat map that incorporates ecological sensitivity.
- Instant alerts containing GPS and fire-risk information.
- Security through authenticated access and authorized recipients.

Limitations:

- Prediction accuracy depends on sensor quality and data availability.
- Harsh weather or sensor failures may affect performance.
- Reliable network connectivity is required for real-time alerts.
- Initial installation and maintenance costs can be high.
- Quantitative field-test results are not provided in the PPT.

4.8 Project Planning, Roles & Budget

Milestone	**Planned Activity**
1	Problem identification and environmental-risk requirements
2	Literature / technical reference review
3	Four-module system and UML design
4	Monitoring, prediction, heat-map and alert implementation
5	Testing and verification
6	Analysis, documentation and presentation

Student	**Assigned Responsibility**	**Actual Contribution**
Chethrika Sri	Module 1 and Module 3	User Authentication & Forest Monitoring; Priority-Based Heat Map Generation; related design, verification and report sections.
Kamali R	Module 2	Forest Fire Prediction; related design, verification and report sections.

Item	**Estimated Cost**	**Actual Cost**
Sensors / GPS hardware	0	0
Software / database resources	0	0
Network / notification services	0	0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The proposed Early Forest Fire Prediction and Alert System with Heat Map-Based Wildlife and Vegetation Protection combines real-time environmental monitoring, early fire-risk prediction, ecological priority mapping and immediate alerting. The design addresses the project problem by linking sensor observations to Low/Medium/High/Critical risk levels, then using wildlife density and vegetation coverage to prioritize response zones. High and Critical conditions are communicated to authorized forest officials with GPS and relevant environmental information. The project supports protection of forests, wildlife and vegetation and explicitly aligns its environmental purpose with SDG 15 – Life on Land. The source presentation does not provide numerical validation, so further field testing is required before claims of measured prediction accuracy or response performance can be made.

5.2 Future Work

- Integrate satellite and drone data for improved detection and situational awareness.
- Use advanced AI models for fire-spread prediction.
- Develop a mobile application for real-time monitoring.
- Integrate automatic emergency response and resource allocation.
- Expand the system to monitor larger forest regions.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Understanding of environmental monitoring requirements for forest-fire prediction.
- Understanding of risk classification and priority-based ecological decision-making.
- Awareness of GPS-based emergency communication and authorized-access requirements.
- Application of UML and modular object-oriented design to an environmental protection system.

Engineering & Professional Skills Developed:

- Requirements analysis and modular system decomposition.
- Risk and trade-off analysis.
- System-level thinking across monitoring, prediction, mapping and alerting.
- Technical documentation, teamwork and engineering communication.

Individual Reflection –
Chethrika Sri (192511112)

Module 1 – User Authentication & Forest Monitoring

As part of Module 1, I worked on the **User Authentication and Forest Monitoring** component of the Early Forest Fire Prediction and Alert System. This module helped me understand the importance of secure access and reliable environmental data collection in a real-time monitoring system. I learned how parameters such as **temperature, smoke, humidity, weather conditions and GPS location** can be continuously monitored and transferred to the fire prediction module. Working on this module improved my understanding of how input data acts as the foundation for the subsequent prediction and alert-generation processes. I also gained knowledge about the importance of authorized access, data reliability and continuous monitoring in an environmental safety application. This work strengthened my understanding of object-oriented analysis and system-level thinking.

Module 3 – Priority-Based Heat Map Generation

In Module 3, I worked on the **Priority-Based Heat Map Generation** component. Through this module, I learned how predicted fire risk can be combined with **wildlife density and vegetation coverage** to identify areas that require different levels of protection. I understood the concept of classifying forest regions into **Priority 1, Priority 2, Priority 3 and Priority 4**, where areas with high fire risk and highly dense wildlife or vegetation receive the highest priority. This module helped me understand how system outputs can be converted into meaningful visual information to support faster decision-making and resource allocation. I also gained a better understanding of how OOAD concepts can be applied to a real-world environmental protection problem.

Overall Reflection

Working on Modules 1 and 3 gave me practical exposure to analyzing a real-world problem and dividing it into functional system components. I improved my understanding of **requirements analysis, system design, environmental data processing, prioritization and UML-based object-oriented design**. The project also made me realize the importance of data accuracy, system reliability, security and sustainability when designing technology for critical applications. Overall, this project improved my technical knowledge as well as my ability to analyze problems, design solutions and contribute effectively to a team-based engineering project.

.

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[1] G. Booch, J. Rumbaugh, and I. Jacobson, The Unified Modeling Language User Guide, 2nd ed. Addison-Wesley Professional, 2005.

[2] R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 9th ed. McGraw-Hill Education, 2019.

[3] Oracle Corporation, Java Platform, Standard Edition Documentation, 2025.

[4] Oracle Corporation, MySQL 8.0 Reference Manual, 2025.

[5] United Nations, Sustainable Development Goal 15: Life on Land, 2025.

Note: These references are taken from the submitted PPT. URLs, DOIs and additional publication metadata were not provided in the source.

APPENDICES

APPENDIX A – DETAILED DESIGN CALCULATIONS / DECISION LOGIC

This appendix documents the decision logic represented in the project presentation. Numerical sensor thresholds are intentionally not assigned because the provided project material does not specify calibrated threshold values.

A.1 Risk Classification Logic

Risk Level	Decision Interpretation	System Response
Low	Environmental readings do not indicate a significant fire condition.	Continue monitoring.
Medium	Moderate abnormal environmental conditions are observed.	Increase attention and continue analysis.
High	Strong abnormal conditions indicate a possible fire event.	Generate an alert and prioritize the affected area.
Critical	Severe conditions indicate an urgent fire threat.	Generate an urgent alert and prioritize immediate response.

A.2 Priority-Based Protection Logic

Priority	Risk / Ecological Condition	Purpose
Priority 1	High fire risk + highly dense wildlife/vegetation	Highest protection priority.
Priority 2	High fire risk + moderate wildlife/vegetation	High protection priority.
Priority 3	Medium fire risk + low wildlife/vegetation	Moderate protection priority.
Priority 4	Low/no fire risk	Safe zone / lowest immediate priority.

A.3 Decision Flow

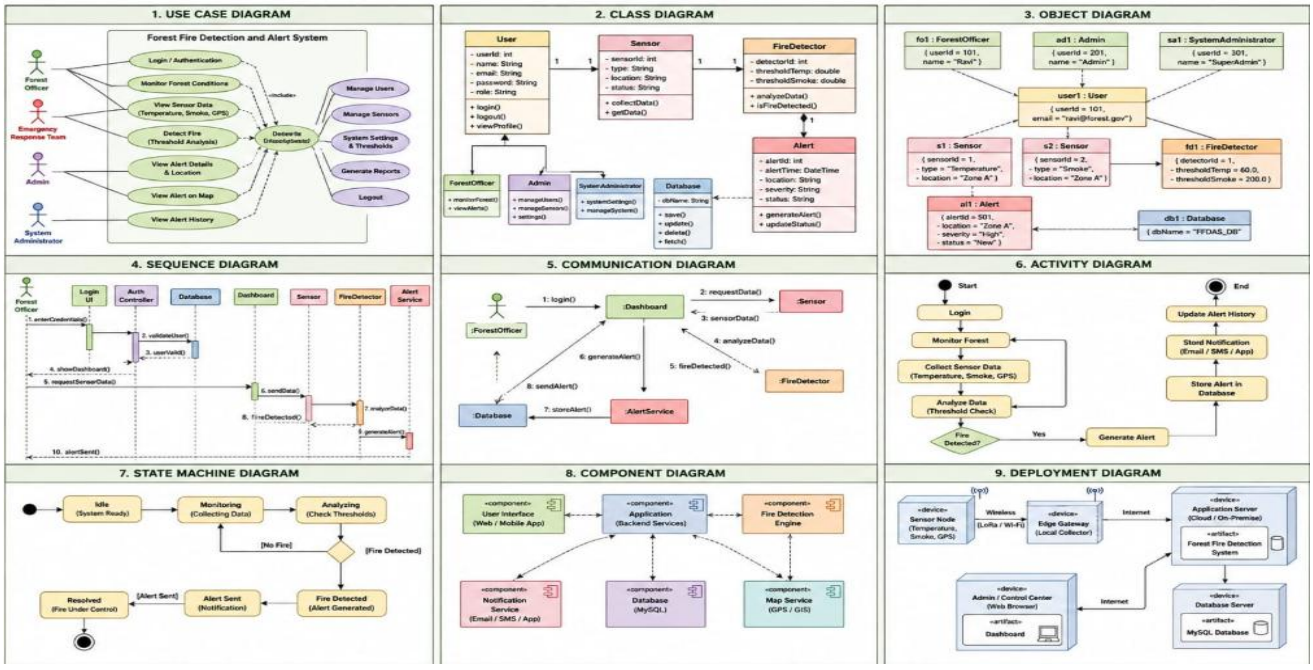
- Authenticate the user before access to monitoring information.
- Collect temperature, smoke, humidity, weather and GPS information.
- Compare current sensor values with predefined thresholds.
- Use historical fire patterns together with current environmental conditions.
- Classify the predicted condition as Low, Medium, High or Critical.
- Combine predicted risk with wildlife density and vegetation coverage.
- Assign Priority 1–4 and update the heat-map representation.
- For High or Critical risk, generate an alert containing GPS location, temperature, smoke, risk, priority and affected area.
- Store alert history for authorized users.

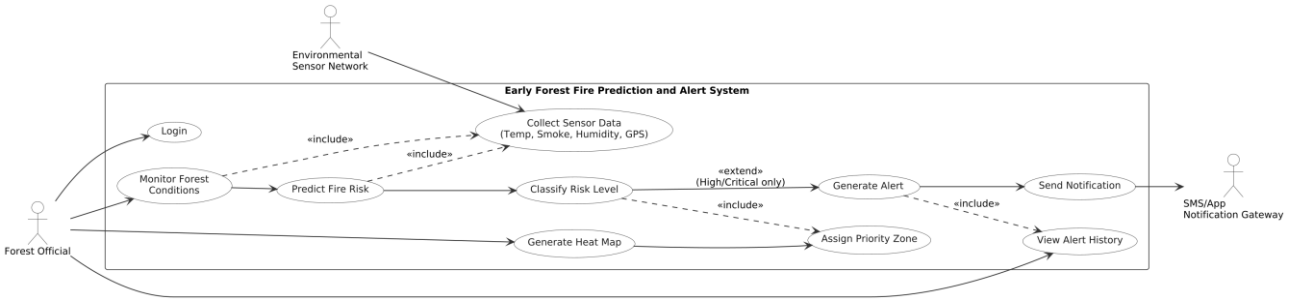
A.4 Reference Pseudocode

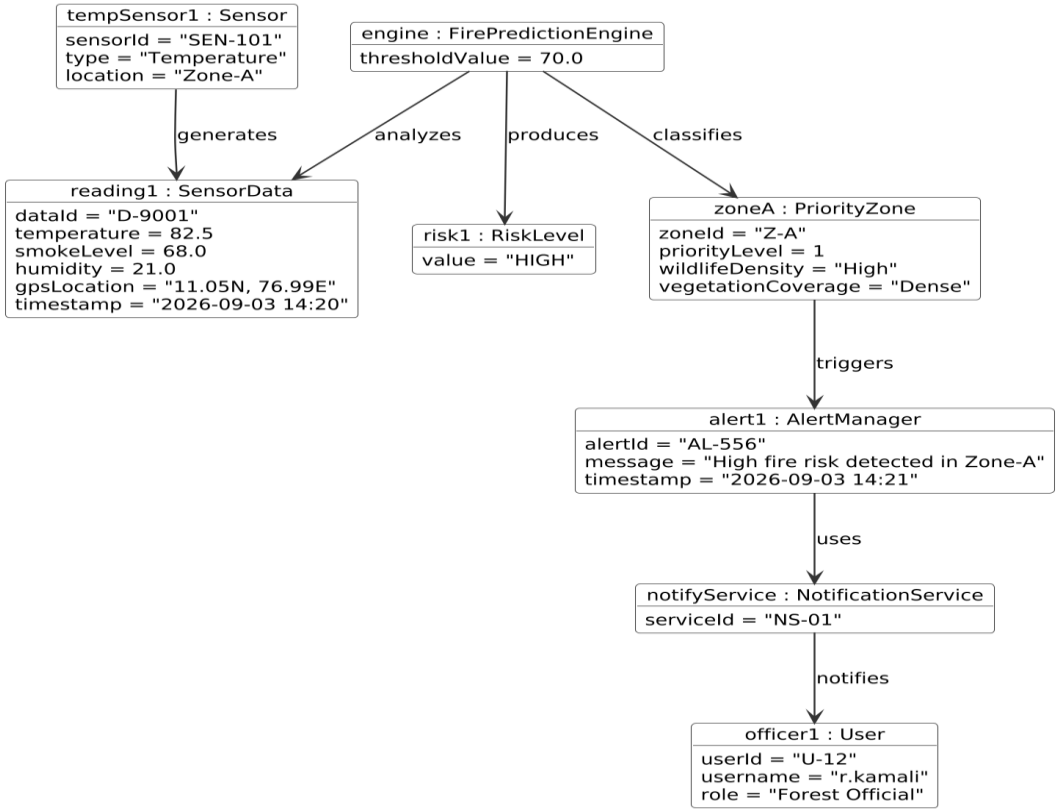
START
Authenticate authorized user
Collect environmental data and GPS
IF data is available THEN
  Compare readings with predefined thresholds
  Analyze historical fire patterns
  Determine risk level: Low / Medium / High / Critical
  Determine ecological priority: 1 / 2 / 3 / 4
  Update heat map
  IF risk is High OR Critical THEN
    Generate alert with location and environmental details
    Record alert in history
  END IF
ELSE
  Report unavailable/incomplete sensor data
END IF
END

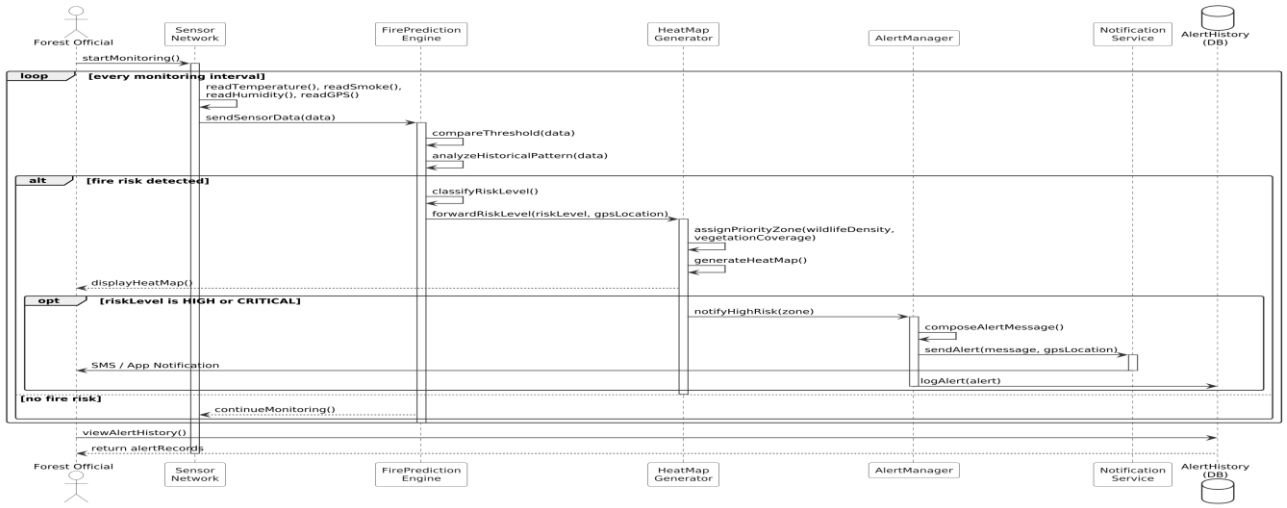
APPENDIX B – ENGINEERING DRAWINGS

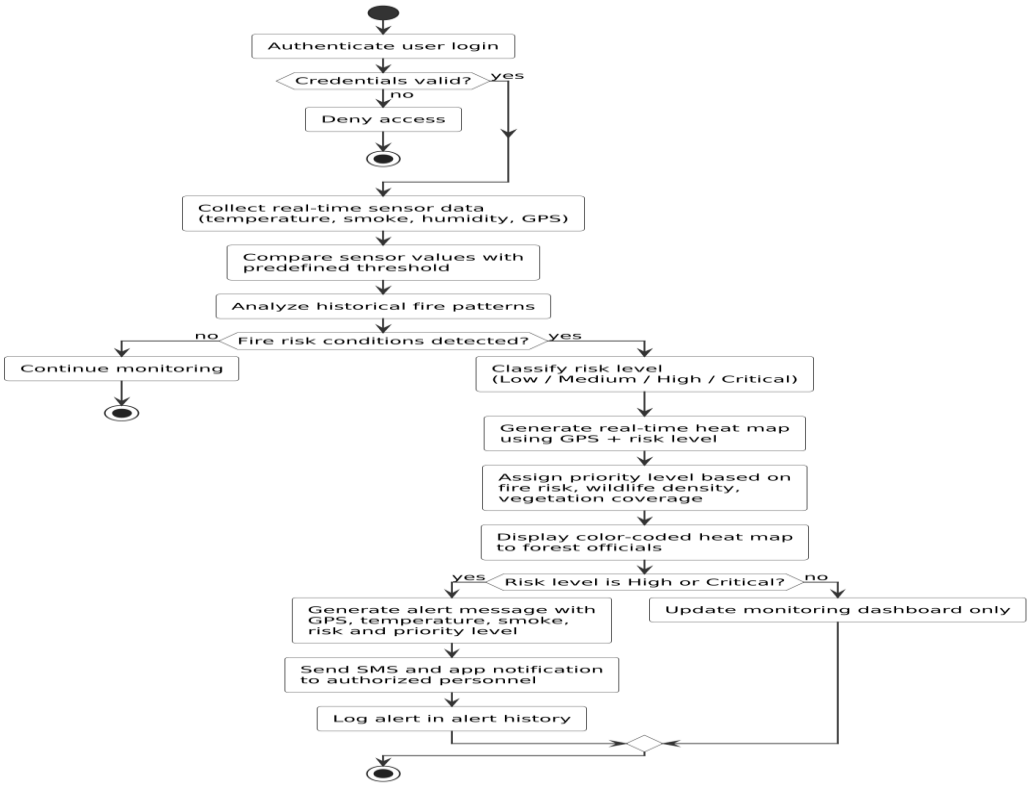
The system architecture and UML representation included in the project presentation form the principal engineering drawings for this report. They describe interaction among authentication, forest monitoring, prediction, priority heat-map generation and alert notification components.

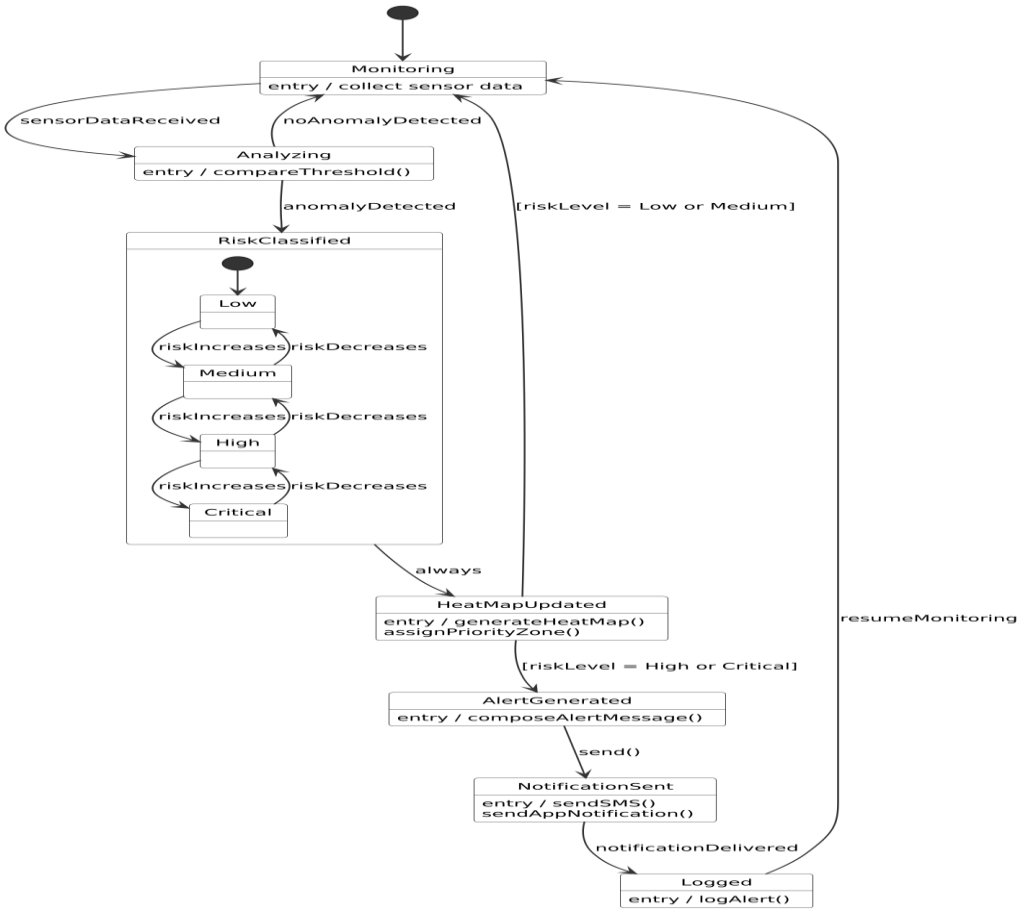


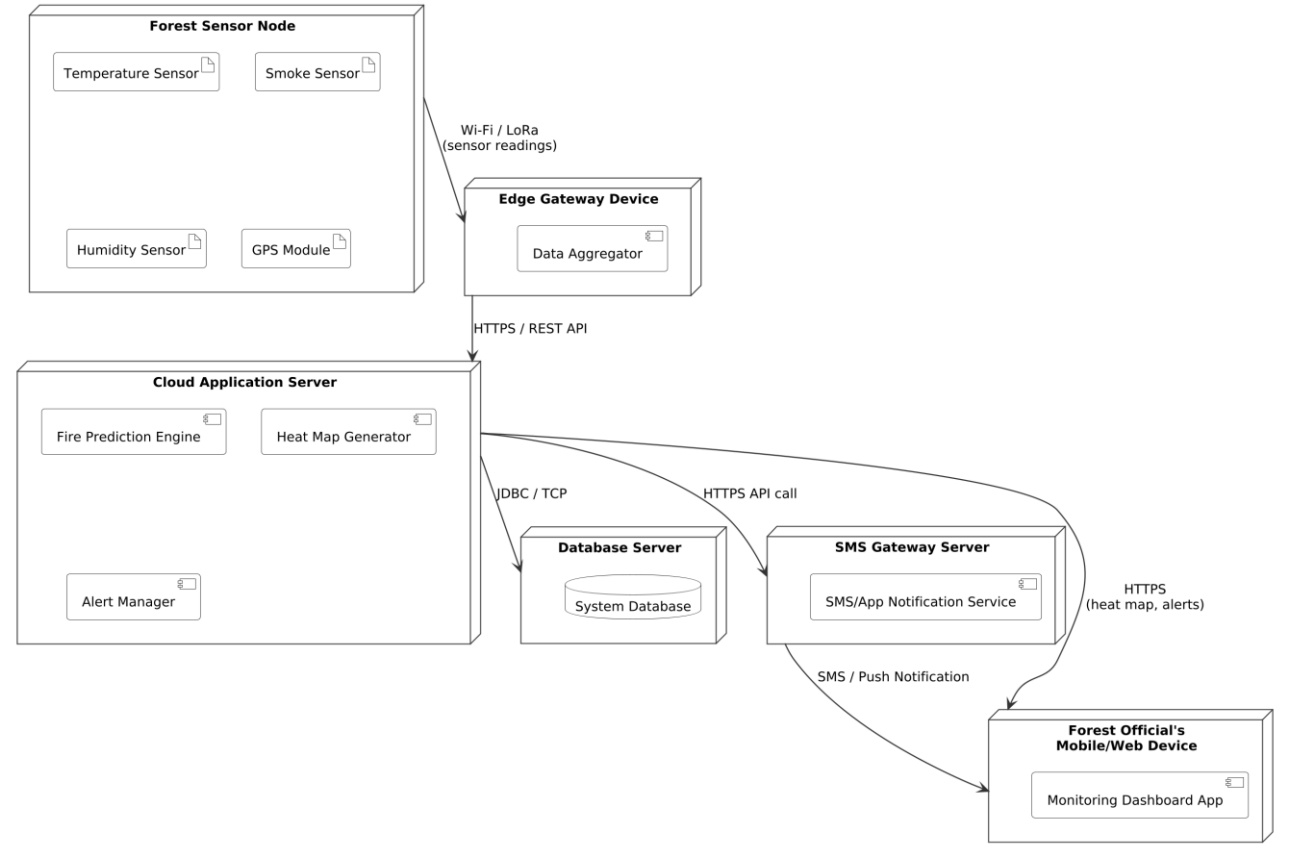

Environmental Sensor Network
Early Forest Fire Prediction and Alert System
Login
Collect Sensor Data (Temp, Smoke, Humidity, GPS)
«include»
«include»
Monitor Forest Conditions
Predict Fire Risk
Classify Risk Level
«extend» (High/Critical only)
Generate Alert
Send Notification
SMS/App Notification Gateway
«include»
«include»
Forest Official
Generate Heat Map
Assign Priority Zone
View Alert History

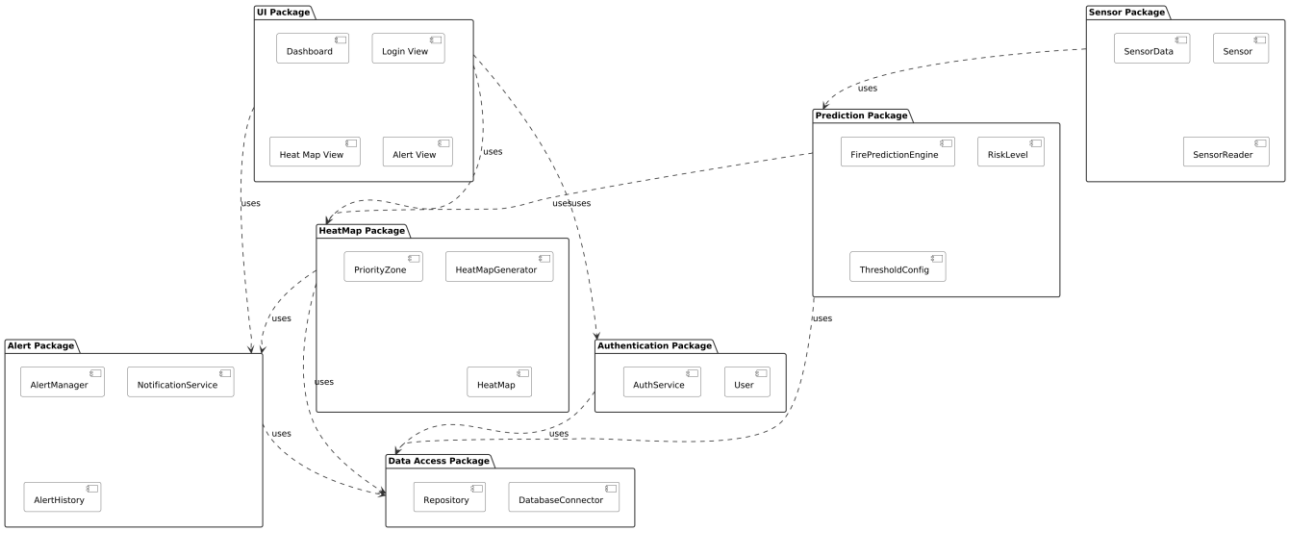

tempSensor1 : Sensor
sensorId = "SEN-101"
type = "Temperature"
location = "Zone-A"
engine : FirePredictionEngine
thresholdValue = 70.0
generates
analyzes
produces
classifies
reading1 : SensorData
dataId = "D-9001"
temperature = 82.5
smokeLevel = 68.0
humidity = 21.0
gpsLocation = "11.05N, 76.99E"
timestamp = "2026-09-03 14:20"
risk1 : RiskLevel
value = "HIGH"
zoneA : PriorityZone
zoneId = "Z-A"
priorityLevel = 1
wildlifeDensity = "High"
vegetationCoverage = "Dense"
triggers
alert1 : AlertManager
alertId = "AL-556"
message = "High fire risk detected in Zone-A"
timestamp = "2026-09-03 14:21"
uses
notifyService : NotificationService
serviceId = "NS-01"
notifies
officer1 : User
userId = "U-12"
username = "r.kamali"
role = "Forest Official"


Forest Official
Sensor Network
FirePrediction Engine
HeatMap Generator
AlertManager
Notification Service
AlertHistory (DB)
startMonitoring()
loop
[every monitoring interval]
readTemperature(), readSmoke(), readHumidity(), readGPS()
sendSensorData(data)
compareThreshold(data)
analyzeHistoricalPattern(data)
alt
[fire risk detected]
classifyRiskLevel()
forwardRiskLevel(riskLevel, gpsLocation)
assignPriorityZone(wildlifeDensity, vegetationCoverage)
generateHeatMap()
displayHeatMap()
opt
[riskLevel is HIGH or CRITICAL]
notifyHighRisk(zone)
composeAlertMessage()
sendAlert(message, gpsLocation)
SMS / App Notification
logAlert(alert)
[no fire risk]
continueMonitoring()
viewAlertHistory()
return alertRecords


Authenticate user login
Credentials valid?
yes
no
Deny access
Collect real-time sensor data (temperature, smoke, humidity, GPS)
Compare sensor values with predefined threshold
Analyze historical fire patterns
Fire risk conditions detected?
no
yes
Continue monitoring
Classify risk level (Low / Medium / High / Critical)
Generate real-time heat map using GPS + risk level
Assign priority level based on fire risk, wildlife density, vegetation coverage
Display color-coded heat map to forest officials
Risk level is High or Critical?
yes
no
Generate alert message with GPS, temperature, smoke, risk and priority level
Update monitoring dashboard only
Send SMS and app notification to authorized personnel
Log alert in alert history


Monitoring
entry / collect sensor data
sensorDataReceived
noAnomalyDetected
Analyzing
entry / compareThreshold()
anomalyDetected
[riskLevel = Low or Medium]
RiskClassified
Low
riskIncreases
riskDecreases
Medium
riskIncreases
riskDecreases
High
riskIncreases
riskDecreases
Critical
always
HeatMapUpdated
entry / generateHeatMap()
assignPriorityZone()
resumeMonitoring
[riskLevel = High or Critical]
AlertGenerated
entry / composeAlertMessage()
send()
NotificationSent
entry / sendSMS()
sendAppNotification()
notificationDelivered
Logged
entry / logAlert()





APPENDIX C – SOURCE CODE / REFERENCE PSEUDOCODE

The project presentation does not provide a complete source-code listing. The following reference pseudocode expresses the proposed functional behavior and is included as design documentation, not as a claim of a completed implementation.

C.1 Forest Fire Prediction

```
FUNCTION predictFireRisk(environmentData, historicalPatterns)
   thresholdResult = compareWithThresholds(environmentData)
   patternResult = analyzeHistoricalPatterns(historicalPatterns, environmentData)

   IF thresholdResult indicates severe abnormality AND patternResult indicates fire possibility
      RETURN "Critical"
   ELSE IF thresholdResult indicates high abnormality
      RETURN "High"
   ELSE IF thresholdResult indicates moderate abnormality
      RETURN "Medium"
   ELSE
      RETURN "Low"
   END IF
END FUNCTION
```

C.2 Priority Assignment

```
FUNCTION assignPriority(riskLevel, wildlifeDensity, vegetationCoverage)
   IF riskLevel == "High" AND wildlifeDensity == "High" AND vegetationCoverage == "High"
      RETURN Priority 1
   ELSE IF riskLevel == "High" AND ecological density is Moderate
      RETURN Priority 2
   ELSE IF riskLevel == "Medium" AND ecological density is Low
      RETURN Priority 3
   ELSE
      RETURN Priority 4
   END IF
END FUNCTION
```

C.3 Alert Generation

```
IF riskLevel == "High" OR riskLevel == "Critical"
   Create alert
   Attach GPS location
   Attach temperature and smoke information
   Attach risk level and priority
   Attach affected-area information
   Send notification to authorized personnel
   Store alert in history
END IF
```

PROJECT REPOSITORY

GitHub Repository: https://github.com/chethrikasri9-bot/CSA1109-OOAD

GitHub Profile: https://github.com/chethrikasri9-bot

The CSA1109-OOAD repository is publicly accessible and contains the project materials and commit history. □cite□turn0view0□

APPENDIX D – SENSOR AND INTERFACE DATASHEET TEMPLATE

Specific sensor manufacturer, model numbers and calibrated specifications were not supplied in the project presentation. The following template records the engineering information to be completed when the physical implementation is finalized.

Parameter	Purpose	Expected Information	Actual Project Specification
Temperature	Detect abnormal heat conditions	Sensor model, range, accuracy, unit, sampling interval	Not specified in PPT
Smoke	Detect smoke-related fire indicators	Sensor model, detection principle, sensitivity, unit	Not specified in PPT
Humidity	Support environmental risk analysis	Sensor model, range, accuracy, sampling interval	Not specified in PPT
Weather	Provide environmental context	Source/sensor, parameters, update interval	Not specified in PPT
GPS	Locate the affected forest area	Module/source, coordinate format, accuracy	Not specified in PPT
Network/Communication	Transmit monitoring and alert information	Technology, range/coverage, availability	Reliable network required; specific technology not specified

Actual sensor datasheets, calibration records and interface specifications should be attached here if they become available.

APPENDIX E – EXPERIMENTAL / RAW DATA AND VERIFICATION RECORD

The project presentation does not contain measured experimental datasets or numerical test results. To avoid presenting invented measurements as project evidence, this appendix provides a structured record for actual data.

E.1 Recommended Raw-Data Format

Timestamp	Temperature	Smoke	Humidity	Weather	GPS	Predicted Risk	Priority	Alert
Actual test record	Enter measured value	Enter measured value	Enter measured value	Enter value	Enter coordinates	Low/Medium/High/Critical	1/2/3/4	Yes/No
Actual test record	Enter measured value	Enter measured value	Enter measured value	Enter value	Enter coordinates	Low/Medium/High/Critical	1/2/3/4	Yes/No

E.2 Verification Test Cases

Test ID	Test Condition	Expected Result	Evidence Status
TC-01	Authorized user enters valid credentials	Access granted to monitoring functions	Implementation evidence not provided in PPT
TC-02	Environmental readings remain within normal range	Low risk / continued monitoring	Implementation evidence not provided in PPT
TC-03	Environmental readings show moderate abnormality	Medium risk classification	Implementation evidence not provided in PPT
TC-04	Environmental readings indicate high fire possibility	High risk + priority assignment + alert	Implementation evidence not provided in PPT
TC-05	Environmental readings indicate critical fire possibility	Critical risk + priority assignment + urgent alert	Implementation evidence not provided in PPT
TC-06	High/Critical event includes GPS and environmental details	Alert contains required location and condition data	Implementation evidence not provided in PPT
TC-07	Alert is generated	Alert is retained in alert history	Implementation evidence not provided in PPT

APPENDIX F – ETHICS, SAFETY AND ENVIRONMENTAL PROTECTION

F.1 Ethical Considerations

- Use the system only for authorized forest-monitoring and emergency-response purposes.
- Protect access to environmental, location and alert information from unauthorized users.
- Clearly distinguish predicted risk from confirmed fire so operators are not misled.
- Maintain traceable alert history so important decisions can be reviewed.
- Use wildlife and vegetation information responsibly, protecting sensitive ecological locations.

F.2 Safety Considerations

- A sensor-based prediction system should support, not replace, trained forest and emergency personnel.
- False negatives may delay response; false positives may consume emergency resources. Thresholds and historical analysis should therefore be validated before operational deployment.
- Sensor failure, harsh weather and network outages must be considered during maintenance and emergency planning.
- Personnel should not enter hazardous areas solely because the system reports a location; normal emergency and forest-safety procedures remain necessary.

F.3 Environmental Considerations

- The main environmental objective is to reduce damage to vegetation, wildlife and biodiversity through earlier detection and prioritization.
- Hardware deployment should minimize physical disturbance to forest ecosystems.
- Future large-scale deployments should consider energy consumption, maintenance access and responsible disposal of electronic components.

APPENDIX G – PROJECT MEETING / PROGRESS RECORD TEMPLATE

Actual meeting minutes were not supplied in the project presentation. The following milestone record is therefore a formal template rather than fabricated meeting evidence.

Stage	Planned Activity	Expected Output	Record / Evidence
1	Problem identification and scope definition	Problem statement, aim and objectives	To be completed from actual project records
2	Requirement analysis	Functional and non-functional requirements	To be completed from actual project records
3	OOAD design	UML and system architecture	Project presentation available

4	Prediction and priority logic	Risk classification and heat-map logic	Project presentation available
5	Alert and notification design	Alert workflow and history concept	Project presentation available
6	Testing and validation	Test results and evidence	Measured evidence not provided in PPT
7	Final documentation	Capstone report and presentation	Current report

APPENDIX H – USER / INDUSTRY FEEDBACK RECORD

No formal user or industry feedback record was supplied in the project presentation. This appendix should be updated if feedback is collected from a faculty reviewer, forest professional, technical mentor or other stakeholder.

Stakeholder	Date	Feedback / Observation	Action Taken	Status
Faculty / Project Guide	Enter date	Enter actual feedback	Enter response/action	Open/Closed
Technical Reviewer	Enter date	Enter actual feedback	Enter response/action	Open/Closed
Potential User / Domain Expert	Enter date	Enter actual feedback	Enter response/action	Open/Closed

APPENDIX I – PROJECT OUTCOMES AND IMPACT

I.1 Sustainable Development Goal Mapping

The project is aligned with United Nations Sustainable Development Goal 15 – Life on Land, as stated in the project presentation. The intended contribution is protection of terrestrial ecosystems, wildlife and vegetation through earlier fire detection and prioritized response.

I.2 Objective-to-Outcome Mapping

Project Objective	Expected Outcome	Evidence Available
Predict forest fires early using real-time environmental data.	Early risk indication using environmental monitoring and prediction logic.	Design described in PPT
Generate a priority-based heat map for wildlife and vegetation protection.	Identification of critical ecological areas for response prioritization.	Design described in PPT
Provide instant GPS alerts to authorized forest officials.	Rapid notification containing location and fire-condition information.	Design described in PPT

I.3 Intended Engineering / Product Outcomes

- Reduced delay between abnormal environmental conditions and awareness of a possible fire.
- Better prioritization of areas where fire risk and ecological value overlap.
- Improved information available to authorized personnel during response.
- A foundation for future integration with satellite, drone, AI-based fire-spread prediction and mobile technologies.

I.4 Current Evidence Status

The supplied presentation establishes the proposed system concept, modules, architecture, UML representation, advantages, limitations and future enhancements. It does not provide a full set of measured performance metrics, deployment records or validated field-test results. Such evidence should be added when available.

APPENDIX J – EVIDENCE



CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Area	Project Contribution	Status / Evidence
Engineering problem definition	Defines the forest-fire detection, prediction, ecological prioritization and alert problem.	Documented
OOAD analysis and design	Uses modular system design, UML and architecture to structure the solution.	Documented in PPT
Requirements and constraints	Considers sensor quality, weather, network availability and cost constraints.	Documented
Implementation / technical feasibility	Defines monitoring, prediction, heat-map and alert workflows.	Concept/design evidence
Testing and verification	Provides structured test cases for future implementation validation.	Test evidence not supplied
Sustainability and ethics	Targets protection of forests, wildlife, vegetation and biodiversity.	Documented
Safety and security	Includes authorized access, alert reliability and operational safety considerations.	Documented
Professional development	Demonstrates application of OOAD concepts to an environmental engineering problem.	Documented
Future innovation	Identifies satellite/drone data, AI fire-spread prediction, mobile app and automated response as enhancements.	Documented

PROJECT REPOSITORY

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GitHub Profile: https://github.com/chethrikasri9-bot

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